

BANKING CUSTOMER & TRANSACTION ANALYSIS REPORT

1. Executive Summary

This project analyzes banking customer, account, and transaction data to identify patterns in transaction activity, account balances, transaction channels, account types, transaction status, customer risk categories, and changes in activity over time. The analysis follows an end-to-end workflow using Excel for initial data cleaning, Python for data validation and exploratory analysis, SQL for business-focused querying, and Power BI for interactive dashboard reporting.

The final dataset contains 34,998 transaction records representing 34,995 unique customers and 34,995 unique accounts. The dataset records approximately ₹4.39 billion in transaction value, with an average transaction amount of about ₹125,370. Completed transactions account for 59.95% of all records.

2. Project Objectives

- Understand the overall customer, account, and transaction profile.
- Analyze transaction value and volume across transaction types and channels.
- Evaluate completed, failed, and pending transaction activity.
- Compare account types and account balances.
- Analyze customer risk-category distribution and transaction behavior.
- Identify monthly transaction trends and changes in activity over time.
- Identify zero-balance transaction patterns across account types.
- Create a professional Power BI dashboard for business reporting.

3. Dataset Overview

The raw Excel dataset contained 35,060 rows and 24 columns. It included customer demographics, account information, transaction details, transaction status, balances, recent transaction activity, and risk classification.

4. Tools & Technologies

Tool	Role	Key Work
Microsoft Excel	Initial cleaning	Duplicates, blanks, currency cleanup, date standardization
Python / Pandas	Validation & analysis	Data types, null/duplicate checks, categorical standardization, EDA
PostgreSQL / SQL	Business analysis	Aggregations, trends, status/channel/risk analysis
Power BI	Visualization	KPI cards, charts, dashboard and interactive reporting

5. Data Preparation & Methodology

5.1 Excel Cleaning

1. Removed 56 duplicate rows from the raw dataset.

2. Filled missing categorical values such as Gender, Occupation, Transaction Channel, Branch, and Risk Category with 'Unknown' where appropriate.
3. Filled missing Annual Income values using the median income of ₹73,077.
4. Cleaned mixed currency formatting in Annual Income, Account Balance, and Transaction Amount.
5. Standardized Account Open Date and Transaction Date using DMY date conversion.
6. Validated age, tenure, balance, transaction amount, and ID fields.

5.2 Python Validation & Exploratory Analysis

- Loaded the cleaned Excel dataset using Pandas.
- Checked shape, data types, missing values, duplicate rows, and descriptive statistics.
- Removed five records containing dates later than the analysis cutoff of 8 October 2026.
- Standardized categorical values such as Account Type, Transaction Status, and Transaction Channel.
- Confirmed zero remaining null values and zero duplicate rows.
- Performed analysis of age, income, balances, transaction amounts, transaction trends, transaction types, account types, risk categories, statuses, and channels.

5.3 SQL Analysis

The final cleaned dataset was loaded into PostgreSQL and analyzed through business-focused SQL queries. The SQL analysis covered overall banking metrics, account type, transaction type, transaction status, transaction channel, account status, monthly trends, risk categories, failed transaction rates, and zero-balance activity.

5.4 Power BI Dashboard

The final CSV dataset was imported into Power BI. Measures were created for key KPIs and the dashboard was designed around transaction performance, status, channels, time trends, account balances, and risk.

6. Data Quality & Validation

Validation	Final Result
Raw records	35,060
Duplicate rows removed	56
Records removed for future dates	5
Final records	34,998
Final columns	24
Remaining null values	0
Remaining duplicate rows	0
Unique customers	34,995
Unique accounts	34,995

7. Key Performance Indicators

KPI	Value
Total Transactions	34,998
Total Customers	34,995
Total Accounts	34,995
Total Transaction Value	₹4,387,691,101.64
Average Transaction Amount	₹125,369.77
Average Account Balance	₹94,381.00
Completed Transactions	20,981
Failed Transactions	7,032
Transaction Success Rate	59.95%

8. Key Analytical Findings

8.1 Transaction Type

Payment transactions generated the highest transaction value at approximately ₹898.36 million, followed by UPI at ₹889.17 million. Payment also had the highest transaction count with 7,132 records. Average transaction amounts were relatively close across all transaction types.

8.2 Transaction Status

Completed transactions represented 59.95% of all transactions. Failed transactions accounted for 20.09%, while pending transactions represented 19.96%. This indicates that approximately 40% of transaction records were not completed at the time represented in the dataset.

8.3 Transaction Channel

UPI generated the highest transaction value among the main channels at approximately ₹886.16 million. ATM, Online Banking, Mobile App, and Branch activity was also closely distributed, with each major channel accounting for roughly one-fifth of transaction records.

8.4 Account Type

Savings accounts were the largest account category with 19,431 accounts and approximately ₹1.67 billion in total balance. Current accounts had the highest average account balance at approximately ₹140,666.

8.5 Account Status

Active accounts represented 60.40% of accounts. Dormant accounts represented 19.92% and closed accounts represented 19.68%. Average balances across the three statuses were relatively similar.

8.6 Risk Category

High, Medium, and Low risk categories were distributed fairly evenly. High-risk records generated the highest transaction value at approximately ₹1.47 billion, but the average transaction and account balance were broadly similar across the three major risk categories.

8.7 Failed Transactions by Channel

UPI had the highest failed transaction rate at 20.42%, followed by Mobile App at 20.20% and ATM at 20.18%. The relatively small difference between channels suggests that failures were not concentrated in one channel.

8.8 Zero-Balance Activity

There were 10,868 records where Balance After Transaction was zero, representing about 31% of all records. Savings accounts had the highest zero-balance rate at 33.54%, while Current accounts had the lowest at 18.34%.

8.9 Monthly Trend

Transaction activity showed a strong overall increase from 2024 through 2026. May 2026 recorded the highest transaction count at 2,376, while July 2026 recorded the highest monthly transaction value at approximately ₹292.25 million. October 2026 was incomplete because the analysis cutoff was 8 October 2026.

9. Power BI Dashboard

The dashboard was designed to provide a compact management view of the banking dataset. It contains four KPI cards and analytical visuals covering transaction value, status, channels, monthly trends, account balances, and risk distribution.

- KPI Cards: Total Transactions, Total Transaction Value, Average Transaction Amount, Transaction Success Rate.
- Transaction Value by Type.

- Transaction Status Distribution.
- Transactions by Channel.
- Monthly Transaction Trend.
- Average Balance by Account Type.
- Transaction Distribution by Risk Category.

10. Business Insights & Recommendations

7. Monitor failed and pending transactions closely because together they represent about 40% of transaction records.
8. Review UPI and other digital channels for failure causes and customer experience improvements.
9. Use the strong growth in monthly transaction activity to support capacity planning and transaction monitoring.
10. Pay additional attention to high-risk customers because the high-risk segment contributes the largest transaction value.
11. Investigate zero-balance patterns, particularly within Savings accounts, to understand whether they represent normal customer behavior or potential service opportunities.
12. Use account-type balance patterns to improve targeted products and customer segmentation.

11. Conclusion

The Banking Customer & Transaction Analysis project demonstrates an end-to-end data analytics workflow, from raw data cleaning and validation to SQL-based business analysis and Power BI visualization. The analysis identifies major transaction patterns, channel behavior, account characteristics, status performance, risk distribution, and time-based trends. The final dashboard converts these findings into a concise format suitable for business reporting and portfolio presentation.